

ACHINTYA RAI

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EDUCATION

University of California, San Diego

Bachelor of Science in Computer Engineering

- Competitive Programming Club, NSF REU Fellow, UC Scholars Research Fellow

Sept. 2024 – June 2028

GPA: 3.9/4.0

WORK EXPERIENCE

Adobe

Software Engineering Intern

San Jose, CA

June 2026 – Sept. 2026

- Profiled **60.6M+** payment events for payments processing, identifying **86,387** duplicates.
- Identified a **26-hour** outage across **1,272** payment segments, establishing the anomaly detector benchmark.
- Built SQL/MLOps pipelines for enterprise data management with weekly retraining and hourly inference.
- Designed ML anomaly detector handling seasonality across **10+** workflows and **30+** countries.

VALT Health

Founding Software Engineer

Remote

Aug. 2025 – Feb. 2026

- Engineered Flask application frameworks ingesting **100k+** datapoints across **20+** databases.
- Accelerated GCP distributed systems batch pipelines by **35%**, cutting database writes by **60%**.
- Automated local setup via Bash scripts, reducing testing time by up to **80%**.

Decklar (formerly Roambee)

Software Engineering Intern

Sunnyvale, CA

June 2024 – Aug. 2024

- Optimized Python pipelines for **1M+** weekly events, achieving a **15%** capacity boost.
- Built data visualization tools for analytics modules, improving reporting speed by **20%**.

RESEARCH

Graph Theory & Neural Architecture

Undergraduate Research Fellow (NSF REU)

San Diego, CA

March 2026 – Present

- Engineered a Python pipeline reducing LLM data footprints from **500GB** to **13GB** manifolds.
- Achieved **30x** runtime reduction enabling large-scale analysis on **8GB RAM** systems.

Applications of Multi-Agent Systems

Undergraduate Research Fellow (UC Scholars)

San Diego, CA

June 2025 – Aug. 2025

Multi-agent AI Chess Tutor | *Django, React, SQLite*

- Built full-stack chess tutor using Django and React application frameworks via Agile sprints.
- Reduced artificial intelligence hallucinations by **80%** across **7** REST API endpoints.

Data Science Teaching & Learning Lab

Undergraduate Researcher

San Diego, CA

Sept. 2024 – Present

ChatSight | *Python, FastAPI, Gemini API*

- Built an artificial intelligence labeling tool processing **16k+** points with **90%** faster throughput.

PROJECTS

SquaredUpChess | *React, Django, AWS, Docker, WebSockets, SQLite*

Aug. 2025 – Dec. 2025

- Engineered payments processing on AWS using Django application frameworks and **20+** REST endpoints.
- Tuned SQLite databases for multi-user scalability, achieving **99.9%** deployment reliability via CI/CD.

NanoGPT | *Python, PyTorch, Transformers, LLM Optimization*

Mar. 2026

- Implemented a GPT-style transformer from scratch, reducing per-token latency by **25%**.

TECHNICAL SKILLS

Languages: Python, SQL (Postgres/SQLite), Java, JavaScript, C/C++, TypeScript, Bash

Frameworks: Django, Flask, React, FastAPI, application frameworks, Docker, PyTest, Asyncio

Data & Cloud: distributed systems, databases, data visualization tools, AWS, GCP, PyTorch, Linux